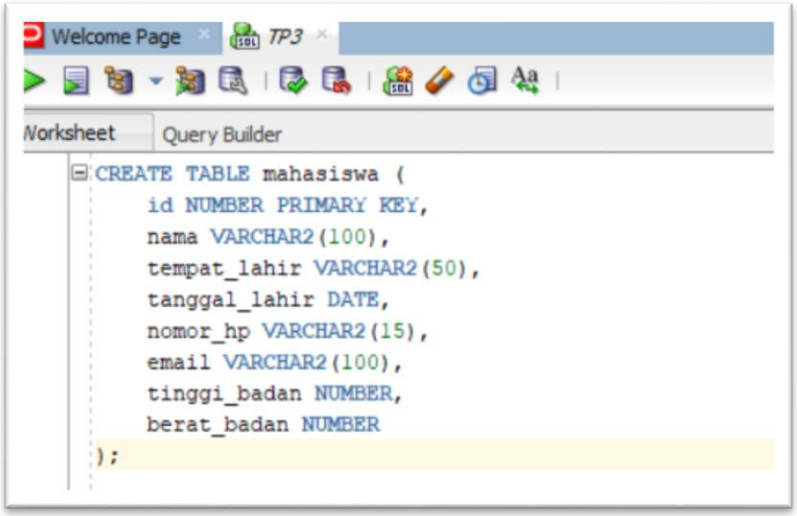
MODUL 3

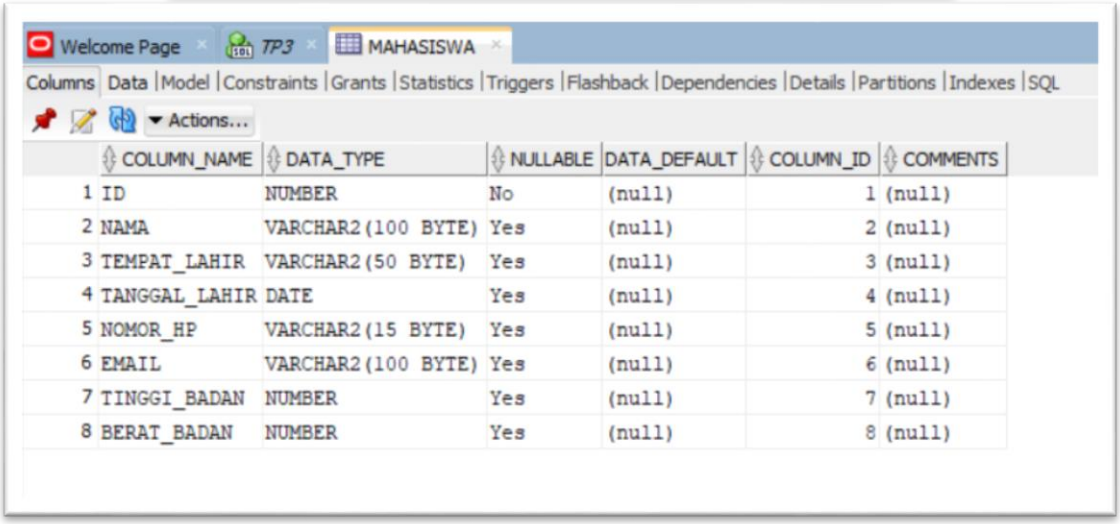
Nama : Khosy Albuchary

NIM : 103122400030

Kelas : SE-08-01

1. Buatlah sebuah table mahasiswa dengan atribut: id, nama, tempat lahir, tanggal lahir, nomor hp, email, tinggi badan, berat badan.





	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID	COMMENTS
1	ID	NUMBER	No	(null)	1	(null)
2	NAMA	VARCHAR2(100 BYTE)	Yes	(null)	2	(null)
3	TEMPAT_LAHIR	VARCHAR2(50 BYTE)	Yes	(null)	3	(null)
4	TANGGAL_LAHIR	DATE	Yes	(null)	4	(null)
5	NOMOR_HP	VARCHAR2(15 BYTE)	Yes	(null)	5	(null)
6	EMAIL	VARCHAR2(100 BYTE)	Yes	(null)	6	(null)
7	TINGGI_BADAN	NUMBER	Yes	(null)	7	(null)
8	BERAT_BADAN	NUMBER	Yes	(null)	8	(null)

2. Lakukan insert data pada table nomor 1 sebanyak 15 rows.

```oracle_sql
INSERT INTO mahasiswa VALUES (1, 'Budi Santoso', 'Jakarta', TO_DATE('15-05-2002', 'DD-MM-YYYY'), '081234567890', 'budi@mail.com', 170.5, 65);
INSERT INTO mahasiswa VALUES (2, 'Ani Wijaya', 'Bandung', TO_DATE('22-08-2003', 'DD-MM-YYYY'), '081234567891', 'ani@mail.com', 160.0, 50);
INSERT INTO mahasiswa VALUES (3, 'Citra Lestari', 'Surabaya', TO_DATE('10-12-2001', 'DD-MM-YYYY'), '081234567892', 'citra@mail.com', 158.2, 48);
INSERT INTO mahasiswa VALUES (4, 'Dedi Kurniawan', 'Medan', TO_DATE('30-01-2002', 'DD-MM-YYYY'), '081234567893', 'dedi@mail.com', 175.0, 70);
INSERT INTO mahasiswa VALUES (5, 'Eka Putri', 'Semarang', TO_DATE('14-03-2003', 'DD-MM-YYYY'), '081234567894', 'eka@mail.com', 162.5, 55);
INSERT INTO mahasiswa VALUES (6, 'Fajar Ramadhan', 'Yogyakarta', TO_DATE('05-11-2002', 'DD-MM-YYYY'), '081234567895', 'fajar@mail.com', 168.0, 60);
INSERT INTO mahasiswa VALUES (7, 'Gita Permata', 'Makassar', TO_DATE('19-07-2004', 'DD-MM-YYYY'), '081234567896', 'gita@mail.com', 155.0, 45);
INSERT INTO mahasiswa VALUES (8, 'Hadi Saputra', 'Palembang', TO_DATE('25-09-2001', 'DD-MM-YYYY'), '081234567897', 'hadi@mail.com', 180.2, 80);
INSERT INTO mahasiswa VALUES (9, 'Indah Sari', 'Malang', TO_DATE('02-06-2002', 'DD-MM-YYYY'), '081234567898', 'indah@mail.com', 165.0, 52);
INSERT INTO mahasiswa VALUES (10, 'Joko Susilo', 'Solo', TO_DATE('12-10-2003', 'DD-MM-YYYY'), '081234567899', 'joko@mail.com', 172.0, 68);
INSERT INTO mahasiswa VALUES (11, 'Kiki Amelia', 'Bogor', TO_DATE('28-04-2002', 'DD-MM-YYYY'), '085612345670', 'kiki@mail.com', 159.5, 49);
INSERT INTO mahasiswa VALUES (12, 'Lutfi Hakim', 'Denpasar', TO_DATE('15-02-2001', 'DD-MM-YYYY'), '085612345671', 'lutfi@mail.com', 178.0, 75);
INSERT INTO mahasiswa VALUES (13, 'Maya Sofia', 'Lombok', TO_DATE('30-12-2003', 'DD-MM-YYYY'), '085612345672', 'maya@mail.com', 161.0, 54);
INSERT INTO mahasiswa VALUES (14, 'Nanda Pratama', 'Padang', TO_DATE('08-08-2002', 'DD-MM-YYYY'), '085612345673', 'nanda@mail.com', 173.5, 66);
INSERT INTO mahasiswa VALUES (15, 'Olivia Rani', 'Pontianak', TO_DATE('20-01-2004', 'DD-MM-YYYY'), '085612345674', 'olivia@mail.com', 164.0, 53);
```

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NOMOR_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN
1	1	Budi Santoso	Jakarta	15-MAY-02	0812345...	budi@mail.com	170.5	65
2	2	Ani Wijaya	Bandung	22-AUG-03	0812345...	ani@mail.com	160	50
3	3	Citra Lestari	Surabaya	10-DEC-01	0812345...	citra@mail.com	158.2	48
4	4	Dedi Kurni...	Medan	30-JAN-02	0812345...	dedi@mail.com	175	70
5	5	Eka Putri	Semarang	14-MAR-03	0812345...	eka@mail.com	162.5	55
6	6	Fajar Rama...	Yogyakarta	05-NOV-02	0812345...	fajar@mail.com	168	60
7	7	Gita Permata	Makassar	19-JUL-04	0812345...	gita@mail.com	155	45
8	8	Hadi Saputra	Palembang	25-SEP-01	0812345...	hadi@mail.com	180.2	80
9	9	Indah Sari	Malang	02-JUN-02	0812345...	indah@mail.com	165	52
10	10	Joko Susilo	Solo	12-OCT-03	0812345...	joko@mail.com	172	68
11	11	Kiki Amelia	Bogor	28-APR-02	0856123...	kiki@mail.com	159.5	49
12	12	Lutfi Hakim	Denpasar	15-FEB-01	0856123...	lutfi@mail.com	178	75
13	13	Maya Sofia	Lombok	30-DEC-03	0856123...	maya@mail.com	161	54
14	14	Nanda Pratama	Padang	08-AUG-02	0856123...	nanda@mail.com	173.5	66
15	15	Olivia Rani	Pontianak	20-JAN-04	0856123...	olivia@mail...	164	53

3. Lakukan proses pemrosesan satu table untuk fungsi-fungsi berikut:

a. Fungsi Karakter dan String (LOWER/UPPER, INITCAP, CONCAT, LENGTH, SUBSTR, LPAD/RPAD, LTRIM, RTRIM, INSTR, REPLACE)

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0.145 seconds

Worksheet | Query Builder

```
SELECT
    LOWER(nama) AS kecil, UPPER(nama) AS BESAR, INITCAP(tempat_lahir) AS Kota,
    CONCAT(nama, ' - Mhs') AS Gabung, LENGTH(nama) AS Pjg, SUBSTR(nama,1,3) AS Sub,
    LPAD(tinggi_badan, 5, '*') AS Lpad, RPAD(berat_badan, 5, '#') AS Rpad,
    INSTR(email, '@') AS Posisi_At, REPLACE(email, 'mail.com', 'id') AS Ganti
FROM mahasiswa;
```

Query Result

All Rows Fetched: 15 in 0.008 seconds

	KECIL	BESAR	KOTA	GABUNG	PJG	SUB	LPAD	RPAD	POSISI_AT	GANTI
1	budi santoso	BUDI SANTOSO	Jakarta	Budi Santoso - Mhs	12	Bud	170.5	65###	5	budi@id
2	ani wijaya	ANI WIJAYA	Bandung	Ani Wijaya - Mhs	10	Ani	**160	50###	4	ani@id
3	citra lestari	CITRA LESTARI	Surabaya	Citra Lestari - Mhs	13	Cit	158.2	48###	6	citra@id
4	dedi kurniawan	DEDI KURNIAWAN	Medan	Dedi Kurniawan - Mhs	14	Ded	**175	70###	5	dedi@id
5	eka putri	EKA PUTRI	Semarang	Eka Putri - Mhs	9	Eka	162.5	55###	4	eka@id
6	fajar ramadhan	FAJAR RAMADHAN	Yogyakarta	Fajar Ramadhan - Mhs	14	Faj	**168	60###	6	fajar@id
7	gita permata	GITA PERMATA	Makassar	Gita Permata - Mhs	12	Git	**155	45###	5	gita@id
8	hadi saputra	HADI SAPUTRA	Palembang	Hadi Saputra - Mhs	12	Had	180.2	80###	5	hadi@id
9	indah sari	INDAH SARI	Malang	Indah Sari - Mhs	10	Ind	**165	52###	6	indah@id
10	joko susilo	JOKO SUSILO	Solo	Joko Susilo - Mhs	11	Jok	**172	68###	5	joko@id
11	kiki amelia	KIKI AMELIA	Bogor	Kiki Amelia - Mhs	11	Kik	159.5	49###	5	kiki@id
12	lutfi hakim	LUTFI HAKIM	Denpasar	Lutfi Hakim - Mhs	11	Lut	**178	75###	6	lutfi@id
13	maya sofia	MAYA SOFIA	Lombok	Maya Sofia - Mhs	10	May	**161	54###	5	maya@id
14	nanda pratama	NANDA PRATAMA	Padang	Nanda Pratama - Mhs	13	Nan	173.5	66###	6	nanda@id
15	olivia rani	OLIVIA RANI	Pontianak	Olivia Rani - Mhs	11	Oli	**164	53###	7	olivia@id

b. Fungsi Tanggal dan Waktu (MM/MON/MONTH, DD/DY, YYYY/YY/RR, AM (or PM)/HH/HH24, MI/SS, ADD_MONTHS(d,n), SYSDATE, LAST_DAY(d), NEXT_DAY(d,day_name), MONTHS_BETWEEN(d1,d2), ROUND(d,[MONTH|YEAR]))

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Worksheet | Query Builder

```
SELECT
    nama,
    TO_CHAR(tanggal_lahir, 'DD-MON-YYYY HH24:MI:SS') AS Full_Date,
    ADD_MONTHS(tanggal_lahir, 6) AS Plus_6_Bln,
    SYSDATE AS Skrg,
    LAST_DAY(tanggal_lahir) AS Akhir_Bln,
    NEXT_DAY(tanggal_lahir, 'MONDAY') AS Next_Mon,
    MONTHS_BETWEEN(SYSDATE, tanggal_lahir) AS Selisih,
    ROUND(tanggal_lahir, 'YEAR') AS Bulat_Thn
FROM mahasiswa;
```

Query Result

SQL | All Rows Fetched: 15 in 0.009 seconds

	NAMA	FULL_DATE	PLUS_6_BLN	SKRG	AKHIR_BLN	NEXT_MON	SELISIH	BULAT_THN
1	Budi Santoso	15-MAY-2002 00:00:00	15-NOV-02	08-MAR-26	31-MAY-02	20-MAY-02	285.80351515830346475507765830346475507 8	01-JAN-02
2	Ani Wijaya	22-AUG-2003 00:00:00	22-FEB-04	08-MAR-26	31-AUG-03	25-AUG-03	270.57770870669056152927120669056152927 1	01-JAN-04
3	Citra Lestari	10-DEC-2001 00:00:00	10-JUN-02	08-MAR-26	31-DEC-01	17-DEC-01	290.96480548088410991636798088410991636 8	01-JAN-02
4	Dedi Kurniawan	30-JAN-2002 00:00:00	30-JUL-02	08-MAR-26	31-JAN-02	04-FEB-02	289.31964419056152927120669056152927120 7	01-JAN-02
5	Eka Putri	14-MAR-2003 00:00:00	14-SEP-03	08-MAR-26	31-MAR-03	17-MAR-03	275.83577322281959378733572281959378733 6	01-JAN-03
6	Fajar Ramadhan	05-NOV-2002 00:00:00	05-MAY-03	08-MAR-26	30-NOV-02	11-NOV-02	280.12609580346475507765830346475507765 8	01-JAN-03
7	Gita Permata	19-JUL-2004 00:00:00	19-JAN-05	08-MAR-26	31-JUL-04	26-JUL-04	259.67448290023894862604540023894862604 5	01-JAN-05
8	Hadi Saputra	25-SEP-2001 00:00:00	25-MAR-02	08-MAR-26	30-SEP-01	01-OCT-01	293.48093451314217443249701314217443249 7	01-JAN-02
9	Indah Sari	02-JUN-2002 00:00:00	02-DEC-02	08-MAR-26	30-JUN-02	03-JUN-02	285.22286999701314217443249701314217443 2	01-JAN-02
10	Joko Susilo	12-OCT-2003 00:00:00	12-APR-04	08-MAR-26	31-OCT-03	13-OCT-03	268.90028935185185185185185185185185185 2	01-JAN-04
11	Kiki Amelia	28-APR-2002 00:00:00	28-OCT-02	08-MAR-26	30-APR-02	29-APR-02	286.38416031959378733572281959378733572 3	01-JAN-02
12	Lutfi Hakim	15-FEB-2001 00:00:00	15-AUG-01	08-MAR-26	28-FEB-01	19-FEB-01	300.80351515830346475507765830346475507 8	01-JAN-01
13	Maya Sofia	30-DEC-2003 00:00:00	30-JUN-04	08-MAR-26	31-DEC-03	05-JAN-04	266.31964419056152927120669056152927120 7	01-JAN-04
14	Nanda Pratama	08-AUG-2002 00:00:00	08-FEB-03	08-MAR-26	31-AUG-02	12-AUG-02	283	01-JAN-03
15	Olivia Rani	20-JAN-2004 00:00:00	20-JUL-04	08-MAR-26	31-JAN-04	26-JAN-04	265.64222483572281959378733572281959378 7	01-JAN-04

c. Fungsi Konversi (TO_NUMBER (char), TO_CHAR (date), TO_CHAR (number), TO_DATE (char))

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Worksheet | Query Builder

```
SELECT
    TO_CHAR(tinggi_badan, '999.99') AS Tinggi_Char,
    TO_CHAR(tanggal_lahir, 'Day, DD Month YYYY') AS Tgl_Char,
    TO_NUMBER('100') + berat_badan AS Tambah_Angka,
    TO_DATE('2023-12-31', 'YYYY-MM-DD') AS Ke_Tgl
FROM mahasiswa;
```

Query Result

SQL | All Rows Fetched: 15 in 0.007 seconds

	TINGGI_CHAR	TGL_CHAR	TAMBAH_ANGKA	KE_TGL
1	170.50	Wednesday, 15 May 2002	165	31-DEC-23
2	160.00	Friday , 22 August 2003	150	31-DEC-23
3	158.20	Monday , 10 December 2001	148	31-DEC-23
4	175.00	Wednesday, 30 January 2002	170	31-DEC-23
5	162.50	Friday , 14 March 2003	155	31-DEC-23
6	168.00	Tuesday , 05 November 2002	160	31-DEC-23
7	155.00	Monday , 19 July 2004	145	31-DEC-23
8	180.20	Tuesday , 25 September 2001	180	31-DEC-23
9	165.00	Sunday , 02 June 2002	152	31-DEC-23
10	172.00	Sunday , 12 October 2003	168	31-DEC-23
11	159.50	Sunday , 28 April 2002	149	31-DEC-23
12	178.00	Thursday , 15 February 2001	175	31-DEC-23
13	161.00	Tuesday , 30 December 2003	154	31-DEC-23
14	173.50	Thursday , 08 August 2002	166	31-DEC-23
15	164.00	Tuesday , 20 January 2004	153	31-DEC-23

d. Fungsi Numerik (ABS(n), MOD (m,n), FLOOR (n), POWER (m,n), ROUND (n[,m]), CEIL (n))

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Worksheet | Query Builder

```
SELECT
    nama, berat_badan,
    ABS(berat_badan - 100) AS Abs_Val,
    MOD(id, 2) AS Sisa_Bagi,
    FLOOR(tinggi_badan) AS Bawah,
    CEIL(tinggi_badan) AS Atas,
    POWER(id, 2) AS Kuadrat,
    ROUND(tinggi_badan, 0) AS Bulat
FROM mahasiswa;
```

Query Result

SQL | All Rows Fetched: 15 in 0.007 seconds

	NAMA	BERAT_BADAN	ABS_VAL	SISA_BAGI	BAWAH	ATAS	KUADRAT	BULAT
1	Budi Santoso	65	35	1	170	171	1	171
2	Ani Wijaya	50	50	0	160	160	4	160
3	Citra Lestari	48	52	1	158	159	9	158
4	Dedi Kurniawan	70	30	0	175	175	16	175
5	Eka Putri	55	45	1	162	163	25	163
6	Fajar Ramadhan	60	40	0	168	168	36	168
7	Gita Permata	45	55	1	155	155	49	155
8	Hadi Saputra	80	20	0	180	181	64	180
9	Indah Sari	52	48	1	165	165	81	165
10	Joko Susilo	68	32	0	172	172	100	172
11	Kiki Amelia	49	51	1	159	160	121	160
12	Lutfi Hakim	75	25	0	178	178	144	178
13	Maya Sofia	54	46	1	161	161	169	161
14	Nanda Pratama	66	34	0	173	174	196	174
15	Olivia Rani	53	47	1	164	164	225	164